

Instructions for AgMIP Low input smallholder system modelling

High information phase

21 September 2018

This phase has two steps, namely (1) calibration and (2) sensitivity analysis. All files can be downloaded at the following address¹:

https://drive.google.com/open?id=1fB_fY7mIRVPR056Jw_LR8kr7z8LDQDaA

This is a new address, please make sure you always use this address.

Please follow carefully the following instructions:

A. Instructions for Calibration

Information on crop management, soil properties and weather were already given during low information phase and are the same. Table 1 gives a brief description of the experiments that were set up during low information phase.

Table 1: Summary of the 10 experiments independently simulated:

Country	Site	Harvest Year	Experiment name	Sowing date	Maturity date	Atmospheric CO2 concentration (ppm) (Keeling curve)*
Mali	MANT	2010	MANT1001	2010-05-31	2010-09-03	389.90
		2009	MANT0901	2009-06-12	2009-09-17	387.36
Ghana	GHKP	2008	GHKP0801	2008-04-20	2008-07-30	385.60
		2009	GHKP0901	2009-04-15	2009-07-25	387.36
Benin	BEOU	2014	BEOU1401	2014-07-04	2014-09-22	398.65
		2015	BEOU1501	2015-06-25	2015-09-19	400.83
Ethiopia	ETBA	2013	ETBA1301	2013-06-07	2013-10-23	396.52
		2014	ETBA1401	2014-06-10	2014-10-27	398.65
Rwanda	RWBU	2014	RWBU1401	2013-09-24	2014-01-22	398.65
		2015	RWBU1501	2014-10-09	2015-02-04	400.83

* Please use this value in your simulations

High information is now provided for each site, for **two growing seasons**, to be used to calibrate your model. We have added the crop and soil information summarized in Table 2.

¹ Please do not share outside the Low-Input smallholder team, see Table 5 for a list of members.

Table 2: New added observed variables for the high information phase with unit, description and timing for the 10 experiments

Variable	Unit	Description	MANT1001	MANT0901	GHKP0801	GHKP0901	BEOU1501	BEOU1401	ETBA1401	ETBA1301	RWBU1501	RWBU1401
Yield	t/ DM/ha	Grain yield	at maturity	at maturity	at maturity	at maturity	at maturity	at maturity	at maturity	at maturity	at maturity	at maturity
Biom	t/ DM/ha	Total above ground biomass	at maturity	at maturity	time series / at maturity	at maturity	time series / at maturity	at maturity	at maturity	at maturity	at maturity	at maturity
LAI	m ² /m ²	In-season leaf area index	time series	time series	time series (1 measurement)	time series (1 measurement)	-	-	time series	-	time series	-
CroN	kg N/ha	Total plant N, cumulative		-	at maturity	at maturity	time series/ at maturity	at maturity	-	-	-	-
CroP	kg P/ha	Total plant P, cumulative		-	at maturity	at maturity	-	-	-	-	-	-
SoilWTot.MaxMeasDepth	mm	Soil water, total in soil profile (to maximum water measurement depth)	time series	-	time series	-	time series	-	time series	time series	time series	time series
SoilNTot.MaxMeasDepth	kg/ha	NO ₃ +NH ₄ N, total in soil profile (up to maximum mineral Nitrogen measurement depth)	-	-	-	-	time series	-	-	-	-	-
SoilW.layer.cm3.cm3	cm ³ /cm ³	Volumetric water content of soil layer	time series	-	time series	-	time series	-	time series	time series	time series	time series
SoilW.layer.mm	mm	Amount of water in soil layer	time series	-	time series	-	time series	-	time series	time series	time series	time series
SoilN.layer.ppm	ppm	NO ₃ +NH ₄ N concentration in soil layer	-	-	-	-	time series	-	-	-	-	-
SoilN.layer.kg.ha	kg[N]/ha	NO ₃ +NH ₄ N amount in soil layer	-	-	-	-	time series	-	-	-	-	-

NB: “-“ means no observation.

The data can be found in the “**Data_Calibration/High information phase/Observations**” folder. The seven files of this folder are described in Table 3.

Table 3: Description of the observations files

File	Description
AgMIP_Smallholders_Observation_daily_average.txt"	Variables time series, average values
AgMIP_Smallholders_Observation_daily_sd.txt"	Variables time series, standard deviation when available (not all averages have an associated standard deviation)
AgMIP_Smallholders_Observation_summary_average.txt"	Summary variables, average values
AgMIP_Smallholders_Observation_summary_sd.txt"	Summary variables, standard deviation when available (not all averages have an associated standard deviation)
AgMIP_Smallholders_Observation_SoilWN_layers_average.txt"	Per layer average soil water and nitrogen time series
AgMIP_Smallholders_Observation_SoilWN_layers_sd.txt"	Per layer soil water and nitrogen standard deviation
AgMIP_Smallholders_Observations_variables_definition.xlsx"	Definition and units of all variables in the above-mentioned files

Please follow these guidelines:

- 1) **Use the new information to improve the calibration of your model.** In the previous phase (Low information), you changed only parameters to adjust time of anthesis and time to maturity. In this phase (**High information**), you can adjust any model parameter to improve your simulations based on the observed data, using whatever technique you normally use. The calibration is site specific, *i.e.* one model calibration per site.
- 2) **Please document the changes you make to your parameterization from the low information to the high information phase in the “Calibration_description.xls” file (“Data_Calibration/High information phase/).** Please make sure you give all the required information (Model code, Modelers name, definition, value of the parameter and calibration method employed for the different sites/phase and general description of your approach). Please rename the file using the Model Code from **Table 5** below. The naming of the result files should be: “**ModelCodeCalibrationDescription.xls**” *e.g.* “STCalibrationDescription.xls” for the STICS model.
- 3) Once the calibration and the simulation runs have been completed, please **save the results into the two templates provided** (folder “**Data_calibration/High information phase**”):
 - **Template_Agmip_Smallholder_Calibration_Summary.txt** is for the SUMMARY information (e.g. final grain yield, final biomass, maximum LAI) – all sites/experiment results must be included in this one file, 1 line per experiment (Five sites with two experiments each = 10 lines, see Table 1). The “Experiment.name” column must contain the experiment names as mentioned in the “field mgmt” sheet of the “field_mgmt_soil_weather.xls” spreadsheet (see also Table 1). Cumulative variables (*i.e.* evaporation, transpiration) need to be computed from sowing to physiological maturity, even if you initiate your model earlier or stop it after maturity. Report in the summary template the cumulative values at maturity.
 - **Template_Agmip_Smallholder_Calibration_Daily.txt** is for the DAILY outputs. Daily results need to be reported from SOWING to MATURITY.
 - Please create one file per experiment (10 files, see Table 1).

- The “experiment name” column must contain the experiment names as mentioned in the “field mgmt” sheet of the “field_mgmt_soil_weather.xls” spreadsheet (see also Table 1).
- The first result line should be the sowing date, no matter if you initiate your model simulation earlier.
- All cumulative variables should be reported starting from “0” at sowing, no matter if you initiate your model earlier.
- Please note that the variables *SoilWTot* and *SoilNTot* need to be computed by integrating all soil layers up to the maximum rooting depth (*SoilWtot.MaxRootDepth* and *SoilNTot.MaxRootDepth*, we want to know what is simulated by the model for the whole profile) and to the maximum sampling depth (*SoilWTot.MaxMeasDepth* and *SoilNTot.MaxMeasDepth*, we want to be able to compare what is simulated to some observations). For each site, these two maximum values are mentioned in the “field_management_soil_weather.xls” file, in the sheet “soils”. When the Maximum measurement depth is set to 0, it means that there were no measurements and you can set the simulated value to “na”.
- Please do not leave black lines after the last simulation day (which should be maturity date).

Description and units for the simulated summary and daily variables are given in the Excel file **Agmip_Smallholder_template_variables_definition.xlsx**.

For each of these two files (summary/daily), please insert in the first column your model name using the Model Code from Table 5. **Please put “na” in the columns corresponding to outputs not simulated by your model and supply all result files in “TAB” limited text files.**

Please rename the template files using the Model Code from Table 5 below. The naming of the result files should be:

- For daily output files (one file for each experiment, i.e. 2 experiment for 5 sites = **total of 10 files**): **ModelCodeExperimentHI.txt**. For example, the experiment name for the first experiment in 2010 from Ntarla, Mali is MANT1001, the result text file of the STICS model for daily outputs for this experiment should be **STMANT1001HI.txt**.
- For the summary output file (**1 file** including all site/experiments): **ModelCodeCalibrationSummaryHI.txt** (e.g. the summary result text file of the STICS model should be **STCalibrationSummaryHI.txt**).

B. Instruction for sensitivity analysis (same instructions as low information phase)

The guidelines for the sensitivity analysis are the same as for low information phase. The sensitivity analysis is performed once the model is calibrated (with high information for this phase).

For each site, the 15 baseline weather files for the sensitivity analysis can be found in the “**Data_Sensitivity_Analysis**” folder. Table 2 shows how climate variable need to be modified to create the different weather files for the sensitivity analysis. Please note that you need to create these files yourself using your own method. The C, T, W variables of the baseline are changed one at a time (addition to daily minimum and maximum temperature, multiplicative factor for daily rainfall, imposed for CO₂ concentration, see Table 2). For each created weather file (30 years), the model needs to be run (independent runs) with three levels of Nitrogen (0, 80, 160 kg/ha split in two applications) each year. Other inputs (soil characteristics, soil initialization, sowing date, fertilization dates, **CO₂ concentration**) must be set to the values corresponding to experiment #1 at each site. In total, there will be 5(sites) * 14(modified climates) * 3(N inputs) * 30(years) + 5(sites) * 1(baseline climate) * 5(N inputs) * 30(years) = 7050 simulations (Table 2).

In Mali, Benin, Ethiopia and Ghana, one growing season correspond to one calendar year, please use 1981 to 2010 (30 years=30 growing seasons). In Rwanda, the growing season extend over two calendar

years. Please use year 1980 to 2010 (31 years, but 30 growing seasons, *i.e* 1980-1981, 1981-1982.... 2008-2009, 2009-2010).

Table 4: List of weather files and N inputs to be used for the sensitivity analysis.

SiteCode refers to the Country(2 letter)Site(2 letters) code, *i.e.* MANT, GHKP, BEOU, RWBU, ETBA.

Change in Climate variable compared with Baseline	Value	Name of Weather file	N management treatments (kg/ha)
CO2 (ppm)	360	SiteCode_C360	0; 80; 160
	450	SiteCode_C450	0; 80; 160
	540	SiteCode_C540	0; 80; 160
	630	SiteCode_C630	0; 80; 160
	720	SiteCode_C720	0; 80; 160
Temperature (°C)	0	SiteCode_T0	0; 80; 160
	+2	SiteCode_T2	0; 80; 160
	+4	SiteCode_T4	0; 80; 160
	+6	SiteCode_T6	0; 80; 160
Water (mm)	50%	SiteCode_W50	0; 80; 160
	75%	SiteCode_W75	0; 80; 160
	100%	SiteCode_W100	0; 80; 160
	125%	SiteCode_W125	0; 80; 160
	150%	SiteCode_W150	0; 80; 160
No change - baseline climate	-	SiteCode_baseline	0;40;80;120;160

Please save the results of simulations into the template provided:

“Template_Agmip_Smallholder_SA.txt”.

Summary results must be included for all the simulations in this one file (7050 lines). Please insert in the first column your model name using the Model Code from Table 3. Please supply all result files in **“TAB”** limited text files. Variable definition for the simulated variables in the Sensitivity template are given in the Excel file **Agmip_Smallholder_template_SA_variables_definition.xlsx**.

Please put **“na”** in the columns corresponding to outputs not simulated by your model.

Please rename the template files using the Model Code from Table 5 below. The naming of the result files should be **(1 file** including all simulations): **ModelCodeSensitivitySummaryHI.txt** (*e.g.* the sensitivity analysis result text file of the STICS model should be **STSensitivitySummaryHI.txt**).

Please return your result files (Calibration : 10 files with daily outputs plus 1 file with annual outputs including all site/years; Sensitivity analysis : 1 file including all virtual experiments, **Calibration description** : 1 file) by **15 January 2019** at the latest to Gatien Falconnier [gatien.falconnier@cirad.fr], with cc to Marc Corbeels [marc.corbeels@cirad.fr].

To facilitate the analysis, we kindly ask you to **fill in the checklist** at the end of this document (Table 6) before sending your results.

Table 5: List of models and modellers for the Agmip Low-Input Smallholder activity

	Model	Model Code	Modelers
1	Agro-IBIS	AG	Tracy Twine
2	APSIM + Option 1	AP1	Peter Thornburn, Elizabeth Meier
3	APSIM + Option 2	AP2	Akinseye Folorunso, Amos Ngwira, Anthony Withbread
4	CELSIUS	CE	François Affholder
5	DSSAT-CERES-Maize+Century	DC1	Cheryl Porter, Gerrit Hoogenboom
6	DSSAT-CERES-Maize+Ceres-SOM	DC2	Ken Boote, Upendra Singh
7	DSSAT-IXIM-Maize+Century	IX1	Jon Lisazo, Ken Boote
8	DSSAT-IXIM-Maize+Ceres-SOM	IX2	Jon Lisazo
9	CROPSYST	CR	Rolf Sommer, Sylvia Nyawira, Claudio Stockle,
10	Daisy	DA	Christian Bugge Henriksen
11	DNDC	DN	Ward Smith, Brian Grant
12	EPIC	EP	Curtis Jones
13	EXPERT-N-Ceres	EC	Sebastian Gayler, Fasil Mequanint
14	EXPERT-N-Spass	ES	Eckart Priesack
15	EXPERT-N-Sucros	EU	Sebastian Gayler, Fasil Mequanint
16	GLAM	GL	Sinha Sumit, Marcelo Galdos, Andrew Challinor
17	GLAM-Parti	GP	Kanhu Pattnayak
18	HERMES	HE	Christian Kersebaum
19	Hybrid-maize	HY	Haishun Yang
20	Info crop	IN	Naresh Kumar Soora
21	MAIZSIM	MA	Dennis Timlin, Soo-Hyung Kim
22	MCWLA - Maize	MC	Tao Fulu, Yi Chen
23	Monica	MO	Claas Nendel, Esther Shupel Ibrahim, Bahareh Kamali
24	PEGASUS	PE	Delphine Deryng
25	RZWQM2	RZ	Laj Ahuja, Saseendran Anapalli, Maha Lotfy Mohamed Elsayed
26	SALUS	SA	Bruno Basso
27	SARRA-H	SR	Christian Barron
28	SIMPLACE-Lintul + Option 1	SI1	Heidi Webber, Babacar Faye
29	SIMPLACE-Lintul + Option 2	SI2	Amit Srivastava, Thomas Gaiser
30	STICS	ST	Patrick Bertuzzi, Eric Justes, Dominique Ripoche, Michel Giner
31	SWB	SW	Michael van de Laan, Isaac Newton Alou

Table 6: Check list for daily, summary and sensitivity analysis result files

	No	Yes
Daily results		
1 Simulation results start line 8		
2 The first simulated day is before or at sowing		
3 The last simulated day is maturity date		
4 File has 16 columns and the order of the variables is the same as in the template		
5 Dates are reported as YYYY-MM-DD		
6 The model name (first column) is your model code as specified in Table 5, please use uppercase		
7 The site (Site.name column) is a 4-letter site code (i.e MANT, GHKP, BEOU, RWBU, ETBA), please use uppercase		
8 The name of the experiment is a 8-characters code as mentionned in Table 1		
9 Missing values are reported as "na"		
10 All values are within expected ranges		
11 All values are given in the correct unit ("Agmip_Smallholder_template_variable_definition.xls"file)		
12 The name of the ten files follow the following scheme ModelCodeExperimentHI.txt		
13 SoilWTot.MaxMeasDepth = SoilWTot.MaxRootDepth in RWBU and BEOU (measurements to the maximum profile depth)		
14 SoilWTot.MaxMeasDepth < SoilWTot.MaxRootDepth in ETBA, MANT and GHKP (measurements shallower than maximum profile depth)		
15 SoilWTot.MaxMeasDepth is "na" in RWBU, ETBA, MANT and GHKP (no soil N measurement)		
Summary results		
16 Simulation results start line 6		
17 File has 15 lines		
18 File has 24 columns and the order of the variables is the same as in the template		
19 Dates (planting date, emergence, anthesis, maturity) are reported as YYYY-MM-DD		
20 The model name (first column) is your model code as specified in Table 5, please use uppercase		
21 The site (Site.name column) is a 4-letter site code (i.e MANT, GHKP, BEOU, RWBU, ETBA), please use uppercase		
22 The name of the experiment is a 8-characters code as mentionned in Table 1, please use uppercase		
23 Missing values are reported as "na"		
24 All values are within expected ranges		
25 All values are given in the correct unit ("Agmip_Smallholder_template_variable_definition.xls"file)		
26 The name of the file follow the following scheme ModelCodeCalibrationSummaryHI.txt		
Sensitivity analysis results		
27 Simulation results start line 7		
28 File has 25 columns and the order of the variables is the same as in the template		
29 Dates (planting date, emergence, anthesis, maturity) are reported as YYYY-MM-DD		
30 The model name (first column) is your model code as specified in Table 5, please use uppercase		
31 The site (Site.name column) is a 4-letter site code (i.e MANT, GHKP, BEOU, RWBU, ETBA), please use uppercase		
32 The weather file (Weather.file) strictly follows the format mentionned in table 4 (e.g. MANT_C360), please use uppercase		
33 There are 30 simulations per Weather variable*Site*N management combinations		